CSA4704 - DEEP LEARNING

CO1 ASSESSMENT 2

Analytical Problem Solving

Analyse the Mathematics of a Diabetic Retinopathy Screening Model

RetinaCare - Mathematical Analysis and Screening Audit

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Task 1 - Break Down the Audit Problem (3 marks)

Question focus: identify the mathematical concept behind each D1-D6 item, link it to the appropriate model stage, and state assumptions and dependencies.

D1-D6 mapping

Data item	Given information	Mathematical concept	Stage
D1	A = [[3, 0], [4, 5]]	Singular value decomposition (SVD); rank-1 approximation	Representation / compression
D2	Quadratic loss, H = [[4, 1], [1, 4]], b = $[5, 5]^T$, initial w = $[0, 0]^T$	Gradient, Hessian, convexity, condition number, gradient descent and momentum	Loss / optimisation
D3	True p = [0.5, 0.25, 0.125, 0.125]; two predicted q distributions	Entropy, cross-entropy, KL divergence	Probability / uncertainty
D4	Prevalence 10%, sensitivity 90%, specificity 95%	Bayes theorem and conditional probability	Decision / screening
D5	True score 10, noise variance 1, Models P and Q predictions	Bias, variance and expected squared error	Prediction error
D6	Confusion matrix counts over 100 images	Accuracy, error rate, precision, recall, specificity and F1	Evaluation

Relationships and assumptions

D1 and D2 address representation and optimisation mathematics; D3 evaluates uncertainty in class probabilities; D4 translates prevalence and test characteristics into post-test probabilities; D5 separates prediction-error components; and D6 assesses observed classification errors.

These are separate numerical examples in the question paper. They should not be treated as measurements from one common trained model unless that relationship is explicitly provided.

- D1 assumes the given 2 × 2 matrix is the feature matrix to compress.
- D2 assumes H is symmetric and the objective is $J(w) = 1/2\ w^T Hw - b^T w$.
- D3 uses log base 2 and the given distributions as the true and predicted class probabilities.
- D4 treats sensitivity as P(positive | disease) and specificity as P(negative | no disease).
- D5 uses the four listed predictions as repeated estimates, with population variance across the four predictions and noise variance $\sigma^2 = 1$.
- D6 takes “referable” as the positive class.

Interpretation for RetinaCare: This decomposition shows how representation, optimisation, uncertainty, decisions and evaluation answer different parts of the RetinaCare audit. The assumptions keep each calculation tied to the data supplied in the paper.

Task 2 - Examine the Compressibility of the Feature Matrix (6 marks)

Given D1: A = [[3, 0], [4, 5]]. The goal is to derive $A = U\Sigma V^T$, compare singular values, calculate a rank-1 reconstruction error and measure retained energy.

Step 1: Calculate A^TA

$$\mathbf{A^T = [[3, 4], [0, 5]]}$$

$$\mathbf{A^TA = [[3, 4], [0, 5]]\ [[3, 0], [4, 5]] = [[25, 20], [20, 25]]}$$

The entries are obtained by row-by-column multiplication: $3^2 + 4^2 = 25$; (3)(0) + (4)(5) = 20; (0)(3) + (5)(4) = 20; and $0^2 + 5^2 = 25$.

Interpretation for RetinaCare: A^TA is symmetric, and its eigenvalues provide the squared singular values of A.

Step 2: Find the eigenvalues

$$\mathbf{\det(A^TA - \lambda I) = 0}$$

$$\mathbf{(25 - \lambda)^2 - 20^2 = 0}$$

$$\mathbf{625 - 50\lambda + \lambda^2 - 400 = 0 \ \Rightarrow\ \lambda^2 - 50\lambda + 225 = 0}$$

$$\mathbf{\lambda = [50 \pm \sqrt{(2500 - 900)}] / 2 = [50 \pm 40] / 2}$$

$$\mathbf{\lambda_1 = 45.0000;\ \lambda_2 = 5.0000}$$

Interpretation for RetinaCare: The eigenvalues differ by a factor of nine, so most squared energy is aligned with the leading feature-space direction.

Step 3: Find the right singular vectors

For $\lambda_1 = 45$: solve $(A^TA - 45I)v = 0$, giving [-20, 20; 20, -20]v = 0 and $v_1 = v_2$. Normalising $[1, 1]^T$ by its length $\sqrt{2}$ gives:

$$\mathbf{v_1 = (1/\sqrt{2})[1, 1]^T}$$

For $\lambda_2 = 5$: solve $(A^TA - 5I)v = 0$, giving [20, 20; 20, 20]v = 0 and $v_1 = -v_2$. Normalising $[1, -1]^T$ gives:

$$\mathbf{v_2 = (1/\sqrt{2})[1, -1]^T}$$

$$\mathbf{V = (1/\sqrt{2})\ [[1, 1], [1, -1]]}$$

Step 4: Find singular values and left singular vectors

$$\mathbf{\sigma_i = \sqrt{\lambda_i} \ \Rightarrow\ \sigma_1 = \sqrt{45} = 3\sqrt{5} = 6.7082;\ \sigma_2 = \sqrt{5} = 2.2361}$$

$$\mathbf{\sigma_1 / \sigma_2 = 6.7082 / 2.2361 = 3.0000}$$

Use $u_i = Av_i/\sigma_i$. For v_1, $Av_1 = (1/\sqrt{2})[3, 9]^T$, so $u_1 = (1/\sqrt{10})[1, 3]^T$. For v_2, $Av_2 = (1/\sqrt{2})[3, -1]^T$, so $u_2 = (1/\sqrt{10})[3, -1]^T$.

$$\mathbf{U = (1/\sqrt{10})\ [[1, 3], [3, -1]]}$$

$$\mathbf{\Sigma = [[3\sqrt{5}, 0], [0, \sqrt{5}]]}$$

$$\mathbf{A = U\Sigma V^T}$$

Approximate matrix forms, to four decimal places:

Matrix	Approximation
U	[[0.3162, 0.9487], [0.9487, -0.3162]]
Σ	[[6.7082, 0.0000], [0.0000, 2.2361]]
V^T	[[0.7071, 0.7071], [0.7071, -0.7071]]

Interpretation for RetinaCare: The SVD represents A as two orthogonal components; the first component has three times the singular value of the second.

Step 5: Rank-1 approximation

$$A_1 = \sigma_1 u_1 v_1^T$$

$$A_1 = (3\sqrt{5})([1, 3]^T/\sqrt{10})([1, 1]/\sqrt{2})$$

$$A_1 = (3/2)\ [[1, 1], [3, 3]] = [[1.5000, 1.5000], [4.5000, 4.5000]]$$

Interpretation for RetinaCare: The rank-1 matrix retains only the leading singular direction and therefore has one nonzero singular value.

Step 6: Frobenius reconstruction error

$$A - A_1 = [[1.5000, -1.5000], [-0.5000, 0.5000]]$$

$$||A - A_1||F = \sqrt{(1.5^2 + (-1.5)^2 + (-0.5)^2 + 0.5^2)}$$

$$||A - A_1||F = \sqrt{5} = 2.2361$$

The same result follows from the SVD property: the optimal rank-1 Frobenius error equals the discarded singular value $\sigma_2 = 2.2361$.

Step 7: Energy retained

$$\text{Energy retained} = \sigma_1^2 / (\sigma_1^2 + \sigma_2^2) \times 100\%$$

$$= 45 / (45 + 5) \times 100\% = 90.0000\%$$

$$\text{Energy discarded} = 100.0000\% - 90.0000\% = 10.0000\%$$

Interpretation for RetinaCare: Rank-1 compression retains 90.0000% of the matrix energy and has an absolute Frobenius error of 2.2361. That is a useful approximation for this small matrix, but clinical adequacy would need to be checked on real retinal images and downstream classification metrics.

Task 3 - Analyse the Curvature of the Loss Surface (5 marks)

Given D2: $J(w) = 1/2\ w^THw - b^Tw$, $H = [[4, 1], [1, 4]]$, $b = [5, 5]^T$, and initial $w = [0, 0]^T$.

Step 1: Eigenvalues and eigenvectors of H

$$\det(H - \lambda I) = (4 - \lambda)^2 - 1 = 0$$

$$(4 - \lambda)^2 = 1 \Rightarrow 4 - \lambda = \pm 1$$

$$\lambda_1 = 5.0000;\ \lambda_2 = 3.0000$$

For $\lambda_1 = 5$, the eigenvector satisfies x = y, so a unit eigenvector is $(1/\sqrt{2})[1, 1]^T$. For $\lambda_2 = 3$, x = -y, giving $(1/\sqrt{2})[1, -1]^T$.

$$\mathbf{v_1 = (1/\sqrt{2})[1, 1]^T;\ \ v_2 = (1/\sqrt{2})[1, -1]^T}$$

Interpretation for RetinaCare: Curvature is greater along v_1 because its eigenvalue is 5; curvature along v_2 is lower, with eigenvalue 3.

Step 2: Positive definiteness and convexity

A real symmetric matrix is positive definite when all its eigenvalues are positive. Here, $\lambda_1 = 5.0000 > 0$ and $\lambda_2 = 3.0000 > 0$, so H is positive definite.

$$\mathbf{det(H) = 4(4) - 1(1) = 15.0000 > 0;\ trace(H) = 8.0000 > 0}$$

Because the Hessian is positive definite, J(w) is strictly convex and has a unique global minimiser.

Interpretation for RetinaCare: The unique minimum means the quadratic objective has no competing local minima.

Step 3: Condition number

$$\mathbf{\kappa_2(H) = \lambda max / \lambda min = 5 / 3 = 1.6667}$$

Interpretation for RetinaCare: The condition number describes the ratio between the largest and smallest curvature. This value indicates a modest curvature difference; gradient descent can converge more evenly than on a much more ill-conditioned surface.

Step 4: Find the minimiser

Set the gradient to zero. Since H is symmetric:

$$\mathbf{\nabla J(w) = Hw - b}$$

$$\mathbf{Hw^* - b = 0 \ \Rightarrow\ w^* = H^{-1}b}$$

$$\mathbf{H^{-1} = (1/15)\ [[4, -1], [-1, 4]]}$$

$$\mathbf{w^* = (1/15)\ [[4, -1], [-1, 4]]\ [5, 5]^T = [1.0000, 1.0000]^T}$$

Step 5: Verify the loss value

$$\mathbf{J(w^*) = 1/2\ [1, 1]\ [[4, 1], [1, 4]]\ [1, 1]^T - [5, 5][1, 1]^T}$$

$$\mathbf{H\ w^* = [5, 5]^T;\ \ w^{*T}Hw^* = 10.0000;\ \ b^Tw^* = 10.0000}$$

$$\mathbf{J(w^*) = 1/2(10.0000) - 10.0000 = -5.0000}$$

Question-paper inconsistency: The task wording asks to verify the minimiser using J(w*) = 0. Under the loss exactly as supplied, J(w*) = -5.0000, not zero. Adding a constant +5 to the objective would shift the minimum value to zero without changing its gradient or minimiser. The calculations here preserve the stated objective.

Interpretation for RetinaCare: The minimum occurs at $w^* = [1.0000, 1.0000]^T$. The value -5.0000 is the minimum of the supplied objective; an additive constant changes the reported objective value but not the optimisation solution.

Task 4 - Differentiate the Models using Information Theory (5 marks)

Given D3: p = [0.5, 0.25, 0.125, 0.125], q_0 = [0.25, 0.25, 0.25, 0.25], and q_1 = [0.4, 0.3, 0.2, 0.1]. Use log base 2.

Step 1: Entropy of the true distribution

$$\mathbf{H(p) = -\Sigma_i\, p_i \log_2(p_i)}$$

$$\mathbf{H(p) = -[0.5(-1) + 0.25(-2) + 0.125(-3) + 0.125(-3)]}$$

$$\mathbf{H(p) = 0.5000 + 0.5000 + 0.3750 + 0.3750 = 1.7500\ bits}$$

Interpretation for RetinaCare: The true distribution contains 1.7500 bits of class uncertainty across the four severity classes.

Step 2: Entropy of each predicted distribution

$$\mathbf{H(q) = -\Sigma_i\, q_i \log_2(q_i)}$$

For the untrained uniform distribution q_0, each class has probability 0.25:

$$\mathbf{H(q_0) = -4(0.25 \log_2 0.25) = 2.0000\ bits}$$

For the partly trained distribution q_1 = [0.4, 0.3, 0.2, 0.1]:

$$\mathbf{H(q_1) = -[0.4 \log_2 0.4 + 0.3 \log_2 0.3 + 0.2 \log_2 0.2 + 0.1 \log_2 0.1]}$$

$$\mathbf{H(q_1) = 1.8464\ bits}$$

Interpretation for RetinaCare: The partly trained model has a less uniform output distribution than the untrained model. Entropy alone does not prove that its probabilities are correct; the distributions must be compared with p.

Step 3: Cross-entropy and KL divergence

$$\mathbf{H(p, q) = -\Sigma_i\, p_i \log_2(q_i)}$$

$$\mathbf{D_{kl}(p \,||\, q) = \Sigma_i\, p_i \log_2(p_i / q_i)}$$

For the untrained model q_0:

$$\mathbf{H(p, q_0) = -\Sigma_i\, p_i \log_2(0.25) = 2.0000\ bits}$$

$$\mathbf{D_{kl}(p \,||\, q_0) = H(p, q_0) - H(p) = 2.0000 - 1.7500 = 0.2500\ bits}$$

For the partly trained model q_1:

$$\mathbf{H(p, q_1) = -[0.5 \log_2 0.4 + 0.25 \log_2 0.3 + 0.125 \log_2 0.2 + 0.125 \log_2 0.1]}$$

$$\mathbf{H(p, q_1) = 1.8007\ bits}$$

$$\mathbf{D_{kl}(p \,||\, q_1) = 1.8007 - 1.7500 = 0.0507\ bits}$$

Quantity (bits)	Untrained q_0	Partly trained q_1
H(p)	1.7500	1.7500
H(q)	2.0000	1.8464
H(p, q)	2.0000	1.8007
$D_{kl}(p \,||\, q)$	0.2500	0.0507

Interpretation for RetinaCare: The partly trained distribution is closer to p in the forward KL direction because its divergence is 0.0507 bits instead of 0.2500 bits.

Step 4: Verify $H(p, q) = H(p) + D_{kl}(p \| q)$

$$\textbf{For } q_0\textbf{: } 1.7500 + 0.2500 = 2.0000 \text{ bits} = H(p, q_0)$$

$$\textbf{For } q_1\textbf{: } 1.7500 + 0.0507 \approx 1.8007 \text{ bits} = H(p, q_1)$$

Step 5: Demonstrate asymmetry and class-wise contributions

$$D_{kl}(q \| p) = \Sigma_i \, q_i \log_2(q_i / p_i)$$

Reverse-direction results: $D_{kl}(q_0 \| p) = 0.2500$ bits and $D_{kl}(q_1 \| p) = 0.0536$ bits. In general, $D_{kl}(p \| q)$ is not equal to $D_{kl}(q \| p)$.

Class	$p_i \log_2(p_i/q_{0i})$	$p_i \log_2(p_i/q_{1i})$	$q_{1i} \log_2(q_{1i}/p_i)$
No DR	0.5000	0.1610	-0.1288
Mild	0.0000	-0.0658	0.0789
Moderate	-0.1250	-0.0848	0.1356
Severe	-0.1250	0.0402	-0.0322

For the untrained reverse direction, the four class-wise contributions $q_{0i} \log_2(q_{0i}/p_i)$ are -0.2500, 0.0000, 0.2500 and 0.2500; these sum to 0.2500 bits.

Individual terms in a KL sum can be negative, but the full KL divergence is non-negative. The divergence is zero exactly when the two distributions are equal on all classes with positive probability.

Why minimising cross-entropy gives the same optimum

$$H(p, q) = H(p) + D_{kl}(p \| q)$$

When p is fixed, H(p) is constant with respect to q. Therefore, minimising H(p, q) over q is equivalent to minimising $D_{kl}(p \| q)$.

Interpretation for RetinaCare: The partly trained q_1 distribution better matches the true distribution than the uniform q_0 distribution, but marginal distributions alone do not determine whether individual patients are classified correctly.

Task 5 - Analyse the Screening Decision using Bayes' Rule (4 marks)

Given D4: prevalence = 10%, sensitivity = 90%, and specificity = 95%. Let D mean referable disease, and + / - denote positive and negative screening results.

Step 1: Write prior and conditional probabilities

$$\textbf{P(D) = 0.1000; P(no D) = 1 - 0.1000 = 0.9000}$$

$$\textbf{P(+ | D) = sensitivity = 0.9000; P(- | D) = 1 - 0.9000 = 0.1000}$$

$$\textbf{P(- | no D) = specificity = 0.9500; P(+ | no D) = 1 - 0.9500 = 0.0500}$$

Step 2: Construct the probability tree quantities

Actual condition	Prior probability	Positive branch	Negative branch
Referable disease	0.1000	0.1000 × 0.9000 = 0.0900	0.1000 × 0.1000 = 0.0100
No referable disease	0.9000	0.9000 × 0.0500 = 0.0450	0.9000 × 0.9500 = 0.8550
Total result probability	1.0000	0.0900 + 0.0450 = 0.1350	0.0100 + 0.8550 = 0.8650

Interpretation for RetinaCare: Out of a population, 13.5000% are expected to screen positive and 86.5000% negative at the supplied prevalence and test characteristics.

Step 3: Probability of disease given a positive result

$$\mathbf{P(D \mid +) = P(+ \mid D)P(D) / P(+)}$$

$$\mathbf{P(D \mid +) = 0.0900 / 0.1350 = 0.6667 = 66.6667\%}$$

Step 4: Probability of disease given a negative result

$$\mathbf{P(D \mid -) = P(- \mid D)P(D) / P(-)}$$

$$\mathbf{P(D \mid -) = 0.0100 / 0.8650 = 0.0116 = 1.1561\%}$$

Interpretation for RetinaCare: A positive result corresponds to a 66.6667% probability of referable disease under these assumptions, while the probability of disease after a negative result is 1.1561%.

Why PPV is not sensitivity; is a positive result sufficient for treatment?

Sensitivity is P(positive | disease): the proportion of diseased patients who test positive. Positive predictive value is P(disease | positive): the proportion of positive patients who actually have disease. These conditional probabilities reverse the conditioning and are not equal in general.

Here, the false-positive share among positive results is 0.0450 / 0.1350 = 0.3333, or 33.3333%. Therefore, a positive screening result alone should not automatically be treated as a definitive diagnosis; clinical examination and confirmatory assessment are needed.

Interpretation for RetinaCare: The posterior probability depends on prevalence as well as test sensitivity and specificity. A screening result informs a clinical decision but is not, by itself, a guaranteed diagnosis.

Task 6 - Trace the Convergence of Gradient Descent (5 marks)

Use D2 with $H = [[4, 1], [1, 4]]$, $b = [5, 5]^T$, initial $w_0 = [0, 0]^T$, and learning rate $\eta = 0.1$.

Step 1: Derive the gradient and update rule

$$\mathbf{J(w) = 1/2\ w^T H w - b^T w}$$

$$\mathbf{\nabla J(w) = Hw - b}$$

$$\mathbf{w_{t+1} = w_t - \eta(Hw_t - b)}$$

The minimiser from Task 3 is $w^* = [1.0000, 1.0000]^T$. Because the two entries remain equal from the zero start, write $w_t = [a_t, a_t]^T$. Then $Hw_t = [5a_t, 5a_t]^T$ and the update becomes $a_{t+1} = a_t - 0.1(5a_t - 5) = 0.5a_t + 0.5$.

Step 2: Perform three iterations

Iteration 1:

$$\mathbf{g_0 = Hw_0 - b = [0, 0]^T - [5, 5]^T = [-5.0000, -5.0000]^T}$$

$$\mathbf{w_1 = [0, 0]^T - 0.1[-5, -5]^T = [0.5000, 0.5000]^T}$$

$$\mathbf{J(w_1) = 5(0.5^2) - 10(0.5) = -3.7500}$$

Iteration 2:

$$\mathbf{g_1 = Hw_1 - b = [2.5000, 2.5000]^T - [5, 5]^T = [-2.5000, -2.5000]^T}$$

$$\mathbf{w_2 = [0.5, 0.5]^T - 0.1[-2.5, -2.5]^T = [0.7500, 0.7500]^T}$$

$$\mathbf{J(w_2) = 5(0.75^2) - 10(0.75) = -4.6875}$$

Iteration 3:

$$\mathbf{g_2 = Hw_2 - b = [3.7500, 3.7500]^T - [5, 5]^T = [-1.2500, -1.2500]^T}$$

$$\mathbf{w_3 = [0.75, 0.75]^T - 0.1[-1.25, -1.25]^T = [0.8750, 0.8750]^T}$$

$$\mathbf{J(w_3) = 5(0.875^2) - 10(0.875) = -4.9219}$$

Step 3: Track distance to the minimiser

$$\mathbf{\|w_t - w^*\|_2 = \sqrt{[(1 - a_t)^2 + (1 - a_t)^2]} = \sqrt{2}\,|1 - a_t|}$$

Point	Weights w	J(w)	Distance to w*
Initial w_0	[0.0000, 0.0000]	0.0000	1.4142
After step 1	[0.5000, 0.5000]	-3.7500	0.7071
After step 2	[0.7500, 0.7500]	-4.6875	0.3536
After step 3	[0.8750, 0.8750]	-4.9219	0.1768

The minimum objective value is -5.0000, so the objective gaps $J(w_t) - J(w^*)$ are 5.0000 initially, then 1.2500, 0.3125 and 0.0781. The gap is multiplied by 0.2500 each iteration along the $\lambda = 5$ eigendirection because the weight error is multiplied by $1 - \eta\lambda = 0.5000$.

Question-paper inconsistency: The loss formula supplied in D2 has minimum -5.0000, not zero. Gradient descent is converging to the minimiser of the stated formula. An additive constant +5 would make the minimum objective value zero without changing the gradient or iterates.

Interpretation for RetinaCare: The weight vector approaches $[1.0000, 1.0000]^T$ and the loss approaches -5.0000. The shrinking distance and objective gap show convergence for $\eta = 0.1$.

Task 7 - Differentiate Momentum from Plain Gradient Descent (4 marks)

Use $\beta = 0.9$ and $\eta = 0.1$ for momentum. Define velocity as $v_t = \beta v_{t-1} + \nabla J(w_t)$ and update $w_{t+1} = w_t - \eta v_t$, with $v_0 = [0, 0]^T$.

Step 1: Two momentum iterations

Iteration 1, starting from $w_0 = [0, 0]^T$:

$$\mathbf{g_0 = Hw_0 - b = [-5.0000, -5.0000]^T}$$

$$\mathbf{v_1 = 0.9[0, 0]^T + [-5, -5]^T = [-5.0000, -5.0000]^T}$$

$$\mathbf{w_1 = [0, 0]^T - 0.1[-5, -5]^T = [0.5000, 0.5000]^T}$$

$$\mathbf{J(w_1) = -3.7500;\ \ ||w_1 - w^*||_2 = 0.7071}$$

Iteration 2:

$$\mathbf{g_1 = Hw_1 - b = [-2.5000, -2.5000]^T}$$

$$\mathbf{v_2 = 0.9[-5, -5]^T + [-2.5, -2.5]^T = [-7.0000, -7.0000]^T}$$

$$\mathbf{w_2 = [0.5, 0.5]^T - 0.1[-7, -7]^T = [1.2000, 1.2000]^T}$$

$$\mathbf{J(w_2) = 5(1.2^2) - 10(1.2) = -4.8000;\ distance = 0.2828}$$

Step 2: Compare with plain gradient descent at $\eta = 0.1$

Method	After 1 update	J after 1	After 2 updates	J after 2
Plain GD	[0.5000, 0.5000]	-3.7500	[0.7500, 0.7500]	-4.6875
Momentum β=0.9	[0.5000, 0.5000]	-3.7500	[1.2000, 1.2000]	-4.8000

Momentum carries part of the previous update into the next step. The second momentum update crosses $w^* = [1, 1]^T$, so it overshoots component-wise, but its Euclidean distance from the optimum is 0.2828 after two steps, less than plain GD's 0.3536 at that point.

Step 3: Learning-rate stability bound

$$\mathbf{For\ plain\ gradient\ descent\ on\ a\ positive\text{-}definite\ quadratic:\ 0 < \eta < 2 / \lambda max(H)}$$

$$\mathbf{\lambda max(H) = 5.0000\ \Rightarrow\ 0 < \eta < 2/5 = 0.4000}$$

For $\eta = 0.5$, plain GD has update factor $1 - \eta\lambda max = 1 - (0.5)(5) = -1.5000$ in the highest-curvature direction. Its error changes sign and grows in magnitude.

$$\mathbf{w_1 = [0, 0]^T - 0.5[-5, -5]^T = [2.5000, 2.5000]^T;\ J(w_1) = 6.2500}$$

$$\mathbf{g_1 = [7.5000, 7.5000]^T;\ w_2 = [2.5, 2.5]^T - 0.5[7.5, 7.5]^T = [-1.2500, -1.2500]^T}$$

$$\mathbf{J(w_2) = 20.3125;\ \ ||w_2 - w^*||_2 = 3.1820}$$

Interpretation of the $\eta = 0.5$ wording: The standard bound above proves failure for plain gradient descent. Under the momentum update defined at the top, the quadratic heavy-ball recurrence has the stability condition $0 < \eta < 2(1 + \beta)/\lambda max = 3.8/5 = 0.7600$ for $\beta = 0.9$. Thus $\eta = 0.5$ is not automatically unstable for that momentum update. The question does not state clearly which

optimiser the $\eta = 0.5$ failure clause refers to; the divergence calculation shown is for plain gradient descent.

Interpretation for RetinaCare: Momentum can accelerate progress but may overshoot. The stability limit depends on the update rule; the $\eta = 0.5$ failure is demonstrated for plain GD, where the rate exceeds 0.4000.

Task 8 - Decompose the Prediction Error (4 marks)

Given D5: true score $y = 10$, noise variance $\sigma^2 = 1$, Model P predictions [8, 9, 8, 9], Model Q predictions [12, 7, 11, 8]. Use population variance across the four listed predictions.

Formulas

$$\textbf{Mean prediction: } \hat{\bar{y}} = (1/m)\,\Sigma_i\, \hat{y}_i$$

$$\textbf{Bias} = \hat{\bar{y}} - y;\ \ \textbf{Bias}^2 = (\hat{\bar{y}} - y)^2$$

$$\textbf{Prediction variance} = (1/m)\,\Sigma_i\, (\hat{y}_i - \hat{\bar{y}})^2$$

$$\textbf{Expected squared error} = \textbf{Bias}^2 + \textbf{Variance} + \sigma^2$$

Model P calculation

$$\hat{\bar{y}}P = (8 + 9 + 8 + 9)/4 = 34/4 = 8.5000$$

$$\text{BiasP} = 8.5000 - 10.0000 = -1.5000;\ \ \text{BiasP}^2 = 2.2500$$

$$\text{VarianceP} = [(-0.5)^2 + (0.5)^2 + (-0.5)^2 + (0.5)^2]/4 = 0.2500$$

$$\text{Expected errorP} = 2.2500 + 0.2500 + 1.0000 = 3.5000$$

Model Q calculation

$$\hat{\bar{y}}Q = (12 + 7 + 11 + 8)/4 = 38/4 = 9.5000$$

$$\text{BiasQ} = 9.5000 - 10.0000 = -0.5000;\ \ \text{BiasQ}^2 = 0.2500$$

$$\text{VarianceQ} = [2.5^2 + (-2.5)^2 + 1.5^2 + (-1.5)^2]/4 = 17/4 = 4.2500$$

$$\text{Expected errorQ} = 0.2500 + 4.2500 + 1.0000 = 5.5000$$

Quantity	Model P	Model Q
Mean prediction	8.5000	9.5000
Bias	-1.5000	-0.5000
Bias squared	2.2500	0.2500
Variance	0.2500	4.2500
Noise variance σ^2	1.0000	1.0000
Expected squared error	3.5000	5.5000

Model P has larger bias but much lower prediction variance. Its expected squared error is lower by 5.5000 - 3.5000 = 2.0000.

Interpretation of the given data: The bias-variance-noise formula is an expected-error decomposition. The empirical mean squared residual of the four listed predictions alone is 2.5000 for P and 4.5000 for Q; these do not include the additional σ^2 term. The task explicitly supplies $\sigma^2 = 1$, so the expected-error values reported above include that noise component.

Interpretation for RetinaCare: Model P is preferable under the requested decomposition because its expected squared error is 3.5000 compared with 5.5000 for Model Q. Model Q has lower squared bias but much higher variance.

Task 9 - Examine the Error Profile of the Classifier (3 marks)

Given D6 confusion matrix: TP = 40, FN = 10, FP = 5, TN = 45, with 100 images total.

Step 1: Accuracy and error rate

$$\textbf{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$\textbf{Accuracy} = (40 + 45)/100 = 0.8500 = 85.0000\%$$

$$\textbf{Error rate} = (FP + FN)/100 = (5 + 10)/100 = 0.1500 = 15.0000\%$$

Step 2: Precision, recall and specificity

$$\textbf{Precision} = TP/(TP + FP) = 40/(40 + 5) = 0.8889$$

$$\textbf{Recall} = TP/(TP + FN) = 40/(40 + 10) = 0.8000$$

$$\textbf{Specificity} = TN/(TN + FP) = 45/(45 + 5) = 0.9000$$

Step 3: F1-score

$$F1 = 2TP/(2TP + FP + FN)$$

$$F1 = 80/(80 + 5 + 10) = 80/95 = 0.8421$$

Metric	Result
Accuracy	0.8500 (85.0000%)
Error rate	0.1500 (15.0000%)
Precision	0.8889 (88.8889%)
Recall / sensitivity	0.8000 (80.0000%)
Specificity	0.9000 (90.0000%)
F1-score	0.8421

There are 10 false negatives and 5 false positives. The false negatives are more frequent and represent referable cases missed by the screening model.

Interpretation for RetinaCare: Accuracy is 85.0000%, but recall is only 80.0000%. Because missed referable disease can delay care, the false-negative count must be considered alongside accuracy and precision.

Task 10 - Relate the Findings to the Audit Question (3 marks)

This task integrates Tasks 2-9 to judge whether the feature compression, loss, optimiser and error profile are appropriate for RetinaCare.

Evidence supporting the mathematical design

- Compression (Task 2): the supplied matrix has singular values 6.7082 and 2.2361. Its rank-1 approximation retains 90.0000% of energy with Frobenius error 2.2361. This supports considering compression, but the matrix is only a small illustrative example.
- Loss curvature (Task 3): H has eigenvalues 5.0000 and 3.0000, is positive definite, and has condition number 1.6667. The quadratic objective has a unique minimiser at $[1.0000, 1.0000]^T$.
- Information theory (Task 4): the partly trained distribution has $D_{kl}(p \parallel q) = 0.0507$ bits, lower than the untrained distribution's 0.2500 bits.
- Optimisation (Tasks 6-7): gradient descent with $\eta = 0.1$ approaches the minimiser; an excessive rate $\eta = 0.5$ makes plain gradient descent diverge for $\lambda max = 5.0000$. Momentum at $\eta = 0.1$ moves to [1.2000, 1.2000] after two updates and overshoots slightly.
- Prediction error (Task 8): Model P has expected squared error 3.5000 versus 5.5000 for Model Q, mainly because Model Q has higher variance.

Concern raised by the screening error profile

Task 9 gives accuracy 0.8500, precision 0.8889, recall 0.8000, specificity 0.9000 and F1 = 0.8421. The classifier misses 10 of 50 actual referable cases in the provided D6 example. That false-negative risk is important clinically.

Integrated conclusion

The mathematical examples support the use of stable optimisation, principled compression and multiple evaluation metrics. However, a mathematically well-conditioned loss or a low reconstruction error does not guarantee a safe screening classifier. The observed false negatives and Task 5 posterior probabilities show that the decision policy must be evaluated with prevalence and clinical error costs in mind.

Interpretation for RetinaCare: RetinaCare should not be audited on accuracy alone. The design decision must combine compression and optimisation evidence with posterior probabilities and the relative consequences of false negatives and false positives.

Task 11 - Analyse the Impact of Low Prevalence (8 marks)

Changed requirement: prevalence falls from 10% to 2%; sensitivity remains 90%, specificity remains 95%; missing a referable case costs ten times as much as an unnecessary referral.

A. Impact analysis (2 marks)

Sensitivity and specificity are conditional test characteristics: sensitivity = P(+ | D) and specificity = P(- | no D). If those conditional values remain 0.9000 and 0.9500, they do not change merely because the

district prevalence changes. The prior P(D), however, changes from 0.1000 to 0.0200, so the posterior probabilities after a positive or negative result change.

$$P(D \mid +) = [Se \times \pi] / [Se \times \pi + (1 - Sp)(1 - \pi)]$$

$$P(D \mid -) = [(1 - Se) \times \pi] / [(1 - Se)\pi + Sp(1 - \pi)]$$

Interpretation for RetinaCare: Prevalence changes the mix of diseased and non-diseased people entering screening, so predictive values change even when sensitivity and specificity stay fixed.

B. Recalculate posteriors and expected counts per 1,000 (2 marks)

First, retain the original 10% prevalence case for comparison. Out of 1,000 people, 100 have disease and 900 do not.

$$TP = 100 \times 0.9000 = 90; \quad FN = 100 \times 0.1000 = 10$$

$$TN = 900 \times 0.9500 = 855; \quad FP = 900 \times 0.0500 = 45$$

$$P(D \mid +) = 90/(90 + 45) = 0.6667$$

$$P(D \mid -) = 10/(10 + 855) = 0.0116$$

Now use 2% prevalence. Out of 1,000 people, 20 have disease and 980 do not.

$$TP = 20 \times 0.9000 = 18; \quad FN = 20 \times 0.1000 = 2$$

$$TN = 980 \times 0.9500 = 931; \quad FP = 980 \times 0.0500 = 49$$

$$P(D \mid +) = 18/(18 + 49) = 18/67 = 0.2687$$

$$P(D \mid -) = 2/(2 + 931) = 2/933 = 0.0021$$

Per 1,000 people	Prevalence 10%	Prevalence 2%
True positives	90	18
False negatives (missed cases)	10	2
True negatives	855	931
False referrals (false positives)	45	49
Positive predictive value	0.6667 (66.6667%)	0.2687 (26.8657%)
P(disease | negative)	0.0116 (1.1561%)	0.0021 (0.2144%)

Interpretation for RetinaCare: At 2% prevalence, a positive result corresponds to only 26.8657% probability of referable disease, even though sensitivity and specificity are unchanged. The count of false referrals rises from 45 to 49 per 1,000, while missed cases fall from 10 to 2 because far fewer people have disease.

C. Analyse policy options and cost (2 marks)

The question asks for the minimum specificity at 2% prevalence required to have $P(D \mid +) \geq 0.5$, assuming sensitivity remains 0.9000 for this calculation.

$$PPV = Se\pi / [Se\pi + (1 - Sp)(1 - \pi)] \geq 0.5$$

For the fraction to be at least 0.5, true positives must be at least as numerous as false positives:

$$Se\pi \geq (1 - Sp)(1 - \pi)$$

$$1 - Sp \leq Se\pi/(1 - \pi) = (0.9000 \times 0.0200)/0.9800$$

$$1 - Sp \leq 0.0183673 \Rightarrow Sp \geq 0.9816327$$

$$\textbf{Minimum specificity} = 98.1633\%$$

At this theoretical operating point, retain sensitivity at 90% and target specificity at 98.1633%. On 1,000 people with 2% prevalence:

$$TP = 18;\ FN = 2;\ FP = 980 \times (1 - 0.9816327) \approx 18;\ TN \approx 962$$

$$PPV = 18/(18 + 18) = 0.5000$$

Let one unnecessary referral cost 1 unit and a missed referable case cost 10 units.

$$\text{Expected cost per 1,000} = 10 \times FN + 1 \times FP$$

$$\text{10\% prevalence, original 90\% / 95\% policy: } 10(10) + 45 = 145 \text{ cost units}$$

$$\text{2\% prevalence, original 90\% / 95\% policy: } 10(2) + 49 = 69 \text{ cost units}$$

$$\text{2\% prevalence, target 90\% / 98.1633\% operating point: } 10(2) + 18 = 38 \text{ cost units}$$

Scenario	FN	FP	Cost = 10FN + FP
10% prevalence, Se 90%, Sp 95%	10	45	145 units / 1,000
2% prevalence, Se 90%, Sp 95%	2	49	69 units / 1,000
2% prevalence, Se 90%, Sp 98.1633% target	2	18	38 units / 1,000

Feasibility caveat: The 98.1633% specificity case is a theoretical comparison that assumes sensitivity can remain at 90.0000%. A real decision threshold moves sensitivity and specificity together along an empirical ROC curve. The question supplies no ROC data for an alternative threshold, so this exact pair cannot be confirmed as attainable from the provided information.

Raising a positive threshold generally increases specificity and reduces false referrals, but can reduce sensitivity and increase missed cases. Lowering the threshold generally increases sensitivity while also increasing false positives. Because a missed referable case costs ten times as much, any threshold change must compare both error types rather than optimise PPV alone.

Interpretation for RetinaCare: Under the fixed-sensitivity calculation, increasing specificity to at least 98.1633% would raise PPV to 0.5000 and reduce the illustrative cost from 69 to 38 units per 1,000 at 2% prevalence. The real threshold must be selected from measured sensitivity-specificity trade-offs because missed cases have a high cost.

D. Evidence-based conclusion (2 marks)

Decision: modify the screening policy and validate a new threshold before deployment. At 2% prevalence, the unchanged 90% sensitivity / 95% specificity setting yields PPV = 0.2687, with 49 false

referrals and 2 missed cases per 1,000 people. The low PPV means most positive results would be false positives, while the cost ratio makes every missed case important.

A target specificity of 98.1633% is the mathematical minimum for PPV ≥ 0.5 if sensitivity remains 90%. This target is not guaranteed to be achievable simultaneously with unchanged sensitivity. RetinaCare should obtain validation data for its score threshold, choose the operating point based on expected clinical cost, and use confirmatory assessment rather than treating a positive screening result as a definitive diagnosis.

> **Interpretation for RetinaCare:** The policy should not be retained unchanged without review. A prevalence-aware threshold and confirmatory pathway are justified, but threshold selection must be evidence-based and must explicitly account for the tenfold cost of a missed referable case.

Final Numerical Answer Sheet

This page consolidates key calculated answers for quick revision. The detailed derivations and interpretations appear in the relevant task sections above.

Task	Key results
2 - SVD	$\lambda = 45.0000, 5.0000$; $\sigma = 6.7082, 2.2361$; rank-1 error = 2.2361; energy retained = 90.0000%.
3 - Hessian	Eigenvalues 5.0000 and 3.0000; $\kappa_2 = 1.6667$; $w^* = [1.0000, 1.0000]^T$; stated $J(w^*) = -5.0000$.
4 - Information theory	$H(p) = 1.7500$ bits; $KL(p\|\|q_0) = 0.2500$; $KL(p\|\|q_1) = 0.0507$; $KL(q_1\|\|p) = 0.0536$.
5 - Bayes	At 10% prevalence: $P(D\|+) = 0.6667$; $P(D\|-) = 0.0116$.
6 - Gradient descent	$w_1 = [0.5000, 0.5000]$, $J = -3.7500$; $w_2 = [0.7500, 0.7500]$, $J = -4.6875$; $w_3 = [0.8750, 0.8750]$, $J = -4.9219$.
7 - Momentum	At $\beta = 0.9$, $\eta = 0.1$: $w_1 = [0.5000, 0.5000]$, $w_2 = [1.2000, 1.2000]$; plain-GD stability bound $\eta < 0.4000$.
8 - Bias / variance	Model P: $\text{bias}^2 = 2.2500$, variance = 0.2500, expected error = 3.5000. Model Q: 0.2500, 4.2500, 5.5000.
9 - Confusion matrix	Accuracy 0.8500; error 0.1500; precision 0.8889; recall 0.8000; specificity 0.9000; F1 0.8421.
11 - Low prevalence	At 2%: PPV = 0.2687, $P(D\|-)=0.0021$, FP = 49 and FN = 2 per 1,000; minimum Sp for PPV 0.5 at Se 0.9 = 98.1633%.